Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/01/2008
PI Name:	Kihm, Kenneth D Ph.D.		
Project Title:	Microscale Investigation of the Thermo-Fluid Transport in the Transition Film Region of an Evaporating Capillary Meniscus Using a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	University of Tennessee		
PI Address 1:	Mechanical, Aerospace and Biomedical Engineering		
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City:	Knoxville	**State:**	TN
Zip Code:	37996	**Congressional District:**	2
Comments:	Moved to U Tennessee from Texas A&M in 2004-2005.		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	10/01/2002	**End Date:**	06/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Son, Sang Young	**Contact Phone:**	216-433-6513
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Flight Program:			
Flight Assignment:	NOTE: NCE goes to 6/30/2008 per PI (4/08) NOTE: Received NCE to 9/30/2007 per PI (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA18G ; NAG3-2712		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In order to enhance the fundamental understanding of thin film evaporation and thereby improve the critical design concept for two-phase heat transfer devices, microscale heat and mass transport is to be investigated for the transition film region using both mathematical modeling and microscale measurement techniques. In addition, the modeling for evaporative liquid metal thin film needs to be carefully examined by redefining the disjoining pressure to account for the unique conditions associated with free electrons. The grant objectives are: 1. Development of a mathematical model to describe the micro/nano-scale fluid flow and heat/mass transfer phenomena in an evaporating extended meniscus, focusing on the transition film region under non-isothermal interfacial conditions. 2. Experimental investigation of thermal transport for evaporating sessile droplets using micro-fabricated heater/detector array. Investigations have been performed for both regular coolant (water) and nanofluidic coolant (water + Al2O3 nanoparticles). 3. Modification and extension of the mathematical model to describe the heat/mass transport for evaporating liquid metal thin film.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advances in medicine and biology hold the promise for improvements in quality of life from a health perspective. Pursuant to this promise is the rapid testing and development of pharmaceuticals. Development of new medicines is increasingly relying upon automated microfluidic technologies such as the lab-on-a-chip. As with microscale heat transfer systems, the role of surface interactions in medical applications is increasing as the length scales decrease. Though the impact of the proposed research on the medical community is far in the future, a more thorough understanding of the transport phenomena in the transition film region will improve the ability to develop new medicines.
Task Progress:	The final progress consists of five primary achievements as described below: I. Effects of the Liquid Polarity on the Heat and Mass Transport Characteristics of the Microscale Evaporating Transition Film A mathematical model is developed to describe the micro/nano-scale fluid flow and heat/mass transfer phenomena in an evaporating extended meniscus, focusing on the transition film region under nonisothermal interfacial conditions. Two different disjoining pressure models, one polar and one non-polar, are considered for water as the working fluid so that the effect of polar interactions between the working fluid and solid surface can be exclusively examined on heat and mass transfer from the thin film. The polarity effect is examined for the thin film established in a 20-micron diameter capillary pore. The results show that for a polar liquid, the transition region of the evaporating meniscus is longer than that of a non-polar liquid. In addition, the strong polar attraction with the solid wall acts to lower the evaporative heat transfer flux. II. Modeling of Evaporating Thin Film Heat/Mass Transport of Binary Mixture Analytical and computational studies were conducted to examine the effect of binary mixture (pentane/decane) on the microscale heat and mass transport of an evaporating meniscus formed inside a two-dimensional slotted pore. Mass conservation in the liquid film is combined with the momentum equations, energy balance, and normal stress balance, and then scaled yielding two constitutive equations: (1) a 4th-order, nonlinear, ordinary differential equation for thin film profile, and (2) a 1st-order, linear, ordinary differential equation for concentration profile. The numerical results showed that the magnitude of distillation-driven capillary stress due to the composition gradient of a binary mixture can be larger than the thermocapillary stress due to temperature gradient while they are acting in opposite direction. Henceforth, the proof-of-concept has been established in that the binary mixture could facilitate improvement of the evaporating thin film stability. It was also shown that the resulting stress elongated the length of the evaporating thin film region without degradation of heat transport effectiveness. III. Spatially and Temporally Resolved Temperature Measurements for Slow Evaporating Sessile Drops Heated by a Microfabricated Heater Array The spatially and temporally resolved evaporation phenomena of a slowly evaporating water droplet are investigated using a micro-fabricated gold heater array consisting of 32 linear heater elements (100 μm wide and 15 mm long, each). Each of the gold micro heater elements works both as a temperature sensor and as a heater. The experiment is performed under a constant voltage mode to examine the spatially resolved temperature history of the droplet contact surface for a period starting at initial contact with the heater and lasting to the point of complete dry-out. The raw data obtained from the linear array have been tomographically deconvolved so that the radial temperature profile can be determined assuming a circular droplet contact surface. IV. Evaporation and Dryout of Nanofluid Droplets on a Microheater Array A microfabricated linear heater array operating at a constant voltage mode has been used to study the effect of nanoparticle size on the evaporation and dryout characteristics of strongly pinned nanofluid droplets. Four different nanofluids have been tested, containing 2-nm Au, 30-nm CuO, 11-nm Al2O3, and 47-nm Al2O3 nanoparticles, each of 5-microliter droplets with 0.5 vol. % in water. Nanofluid droplets show strong pinning along the droplet perimeter and, upon evaporation, leave a ring-shaped nanoparticle stain. Particle size is seen to have a clear and strong effect on the dryout stain pattern, while heater temperature seems to have little effect. With the assumption of axi-symmetry, tomographic deconvolution of measured data from the linear heater array allows for examination of the spatially and temporally resolved temperature and heat flux characteristics of the evaporating nanofluid droplets. V. Modeling Alkaline Liquid Metal Evaporating Thin Films Using Both Retarded Dispersion and Electronic Force Components A new thin film evaporation model is presented that captures the unsimplified dispersion force along with an electronic disjoining pressure component that is unique to liquid metals. The resulting nonlinear 4th-order ODE is solved using implicit orthogonal collocation along with the Levenberg-Marquardt method. The electronic component of the

	disjoining pressure should be considered when modeling liquid metal extended meniscus evaporation for a wide range of work function boundary values, which represent physical properties of different liquid metals. For liquid sodium, as an example test material, variation in the work function produces order-of-magnitude differences in the film thickness and evaporation profile.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Wee SK, Kihm KD, Hallinan K. "Effects of the liquid polarity and the wall slip on the heat and mass transport characteristics of the micro-scale evaporating transition film." International Journal of Heat and Mass Transfer, 2005 Jan;48(2):265-78. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2004.08.021 , Jan-2005
Articles in Peer-reviewed Journals	Wee SK, Kihm KD, Pratt DM, Allen JS. "Microscale heat and mass transport of evaporating thin film of binary mixture." Journal of Thermophysics and Heat Transfer. 2006 Apr-Jun;20(2):320-6. , Apr-2006
Articles in Peer-reviewed Journals	Chon CH, Paik SW, Tipton JB Jr, Kihm KD. "Evaporation and dryout of nanofluid droplets on a microheater array." Journal of Heat Transfer. 2006 Aug;128(8):735. http://dx.doi.org/10.1115/1.2221298 , Aug-2006
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Articles in Peer-reviewed Journals	Chon CH, Paik S, Tipton JB Jr, Kihm KD. "Effect of nanoparticle sizes and number densities on the evaporation and dryout characteristics for strongly pinned nanofluid droplets." Langmuir. 2007 Mar 13;23(6):2953-60. PMID: 17338500 , Mar-2007
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Dissertations and Theses	Tipton JB Jr. "Modeling Alkaline Liquid Metal Evaporating Thin Films Using Both Retarded Dispersion and Electronic Force Components." [Ph.D. Dissertation], University of Tennessee, Mechanical, Aerospace and Biomedical Engineering Department (expected to graduate in 2009). , Jan-2009
Dissertations and Theses	Chon CH. "The Thermophysical Role of Nanoparticles in Nanofluidic Heat and Mass Transport." Ph.D. Dissertation, University of Tennessee, Mechanical, Aerospace and Biomedical Engineering Department, 2006. , May-2006
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